LEARNER GUIDE

Perimeter or area?

Perimeter and Area · Lesson 1. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

The **perimeter** of a shape is the distance all the way round its edge, like a fence. It is a length, measured in metres (m). The **area** is the space inside, like the grass. It is counted in squares: a square 1 m by 1 m is **1 square metre (1 m^2)**. Perimeter and area are different stories: 20 m of fence can hold anything from 9 m^2 to 25 m^2 of grass, depending on the shape.

What you are learning to notice

- Perimeter goes round the edge (m); area fills the inside (m^2).
- Shapes with the same perimeter can have different areas.
- Shapes with the same area can have different perimeters.
- For a fixed fence, the square holds the most grass.
- Always give the unit: m for a length, m^2 for an area.

Moving through it, one step at a time

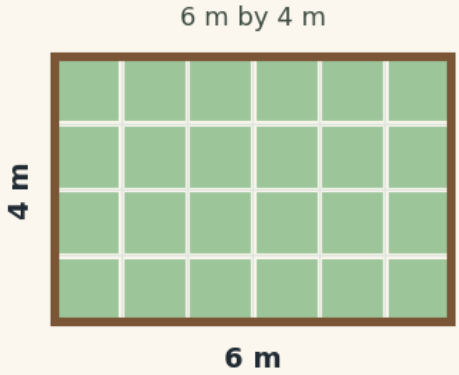
For the perimeter of a rectangle, add all four sides: length + width + length + width. For its area, count the 1 m squares inside: the number of rows times the number in each row. Then check the units: perimeter in m, area in m^2.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Fence and grass

Find the perimeter and the area of this plot.

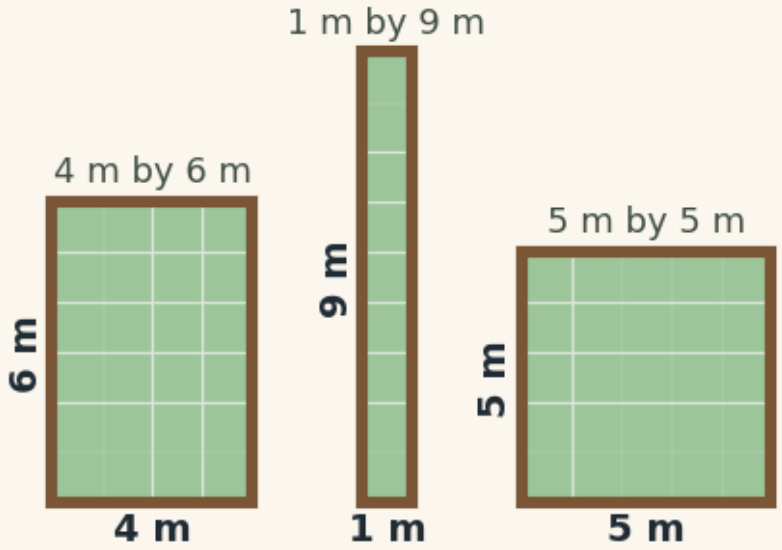


1. Perimeter: add all four sides.	6 + 4 + 6 + 4 = 20 m
2. Area: count the squares inside.	4 rows of 6 = 24 squares
3. Each square is 1 m^2, so	the area is 24 m^2

Perimeter 20 m; area 24 m^2.

Example 2 — Same fence, different grass

Each plot uses 20 m of fence. How much grass does each hold?



1. 4 m by 6 m: 4 + 6 + 4 + 6 = 20 m of fence. 24 m^2 of grass

2. 1 m by 9 m: 1 + 9 + 1 + 9 = 20 m of fence. 9 m^2 of grass

3. 5 m by 5 m: 5 + 5 + 5 + 5 = 20 m of fence. 25 m^2 of grass

Same perimeter, different areas. The square holds the most.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (counting squares, and metres (Place Value Lesson 8)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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